End-to-End Machine Learning Project

CSCI-P 556

ZORAN TIGANJ

Reminders/Announcements

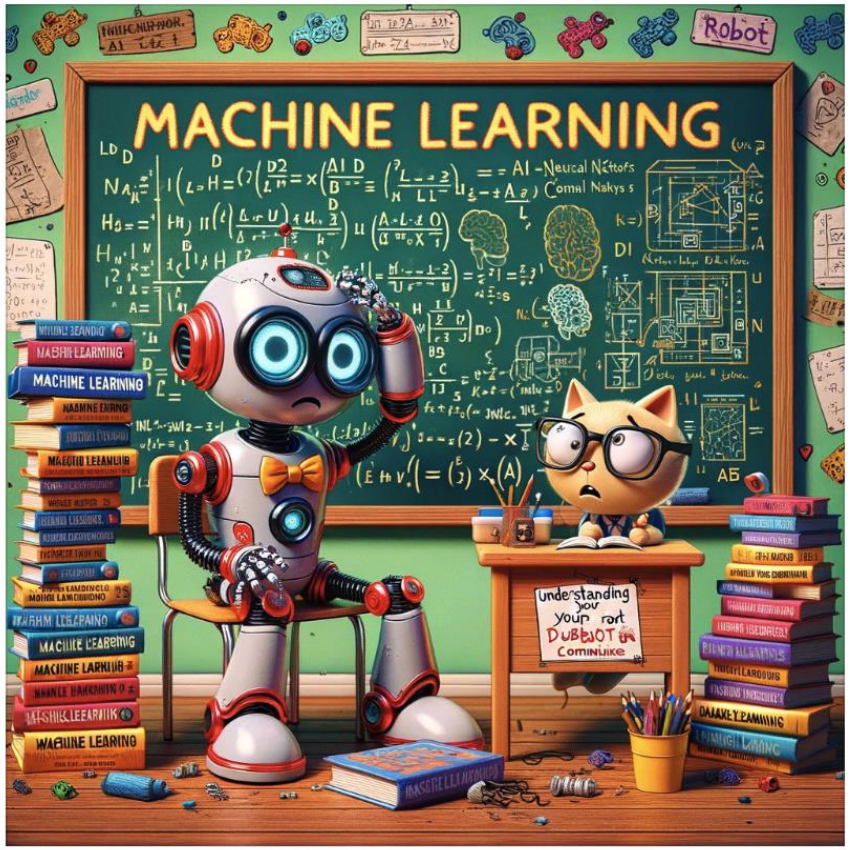
- Quiz deadline today (this will be a general pattern – we will have a weekly quiz with a deadline on Wednesday).
- Dates for all assignments, midterm and final exams are on Canvas.
  - Midterm exam: October 13 or October 20 (let me know if you won't be able to make it on either of these days and we will go with a date that has fewer conflicts)
  - Final exam: Wednesday, December 17 from 12:40-2:40 p.m? (This is per official IU schedule, last call to let me know if you won't be in town at this time)
  - Office hours starting this week (see Canvas for up to date schedule).

ChatGPT 4

You

Create an example of a funny digital art about machine learning

ChatGPT



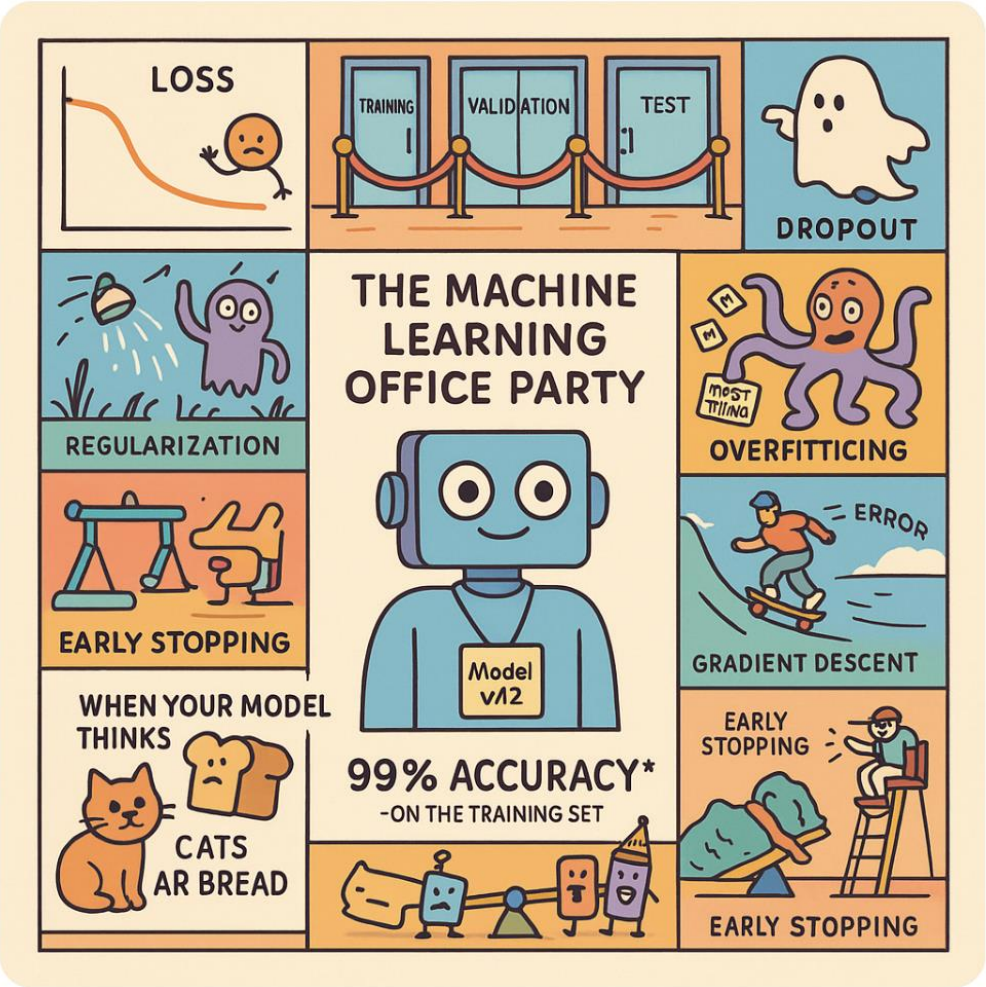
Here's a funny digital art about machine learning. It features a humorous scene with a robot and a cat in a classroom setting, surrounded by books and a blackboard filled with equations. Enjoy the playful and colorful depiction!

ChatGPT 5 Thinking

Create an example of a funny digital art about machine learning

Thought for 16s

Image created



End-to-End Machine Learning Project

- 1. Look at the big picture.
- 2. Get the data.
- 3. Discover and visualize the data to gain insights.
- 4. Prepare the data for Machine Learning algorithms.
- 5. Select a model and train it.
- 6. Fine-tune your model.
- 7. Present your solution.
- 8. Launch, monitor, and maintain your system.

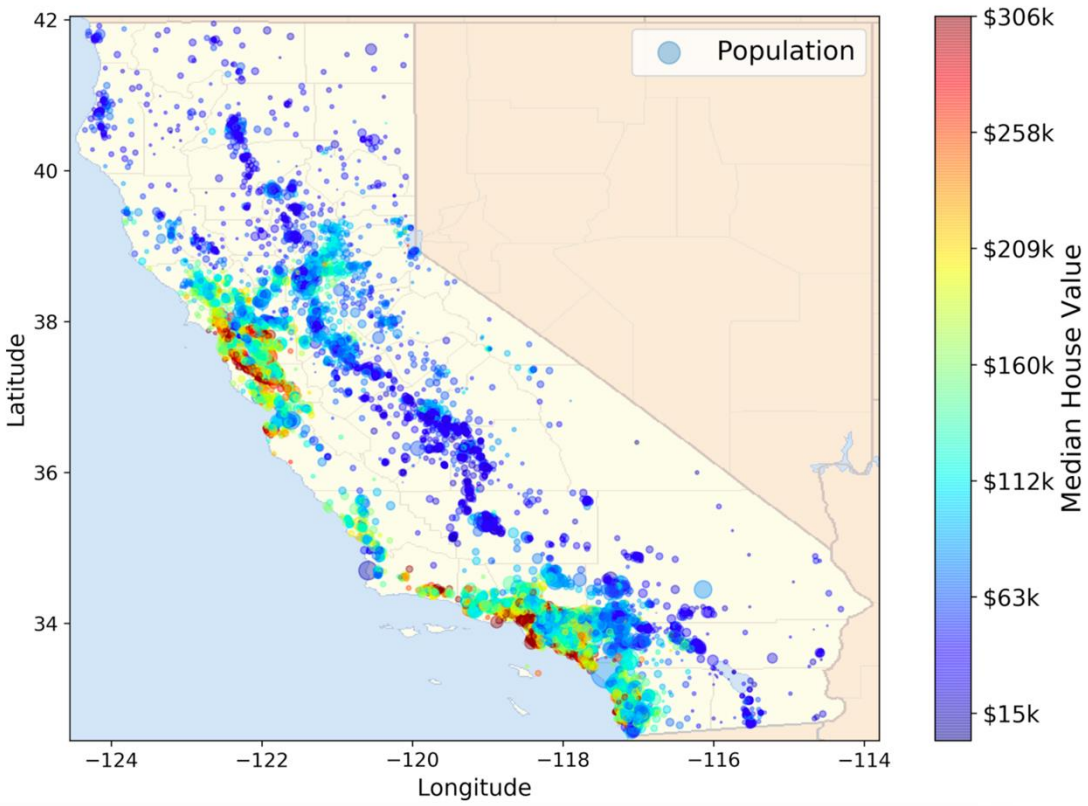
The lecture follows chapter 2 of the *Hands on Machine Learning* textbook

Where to find datasets

- There are thousands of open datasets to choose from, ranging across all sorts of domains.
- Popular open data repositories
  - UC Irvine Machine Learning Repository
  - Kaggle datasets
  - Amazon's AWS datasets
  - Tensorflow datasets
- Meta portals (they list open data repositories)
  - Data Portals
  - OpenDataMonitor
  - Quandl
- Other pages listing many popular open data repositories
  - Wikipedia's list of Machine Learning datasets
  - Quora.com
  - The datasets subreddit

California Housing Prices dataset

- We'll use the California Housing Prices dataset from the StatLib repository.
- This data includes metrics such as:
  - population,
  - median income,
  - median housing price for each block group (district) in California.
- Our model should learn from this data and be able to predict the median housing price in any district, given all the other metrics.



A machine learning pipeline for real estate investments

- What's the objective:
  - Predict district's median housing price so your company can decide whether to invest by buying a property in a given area or not

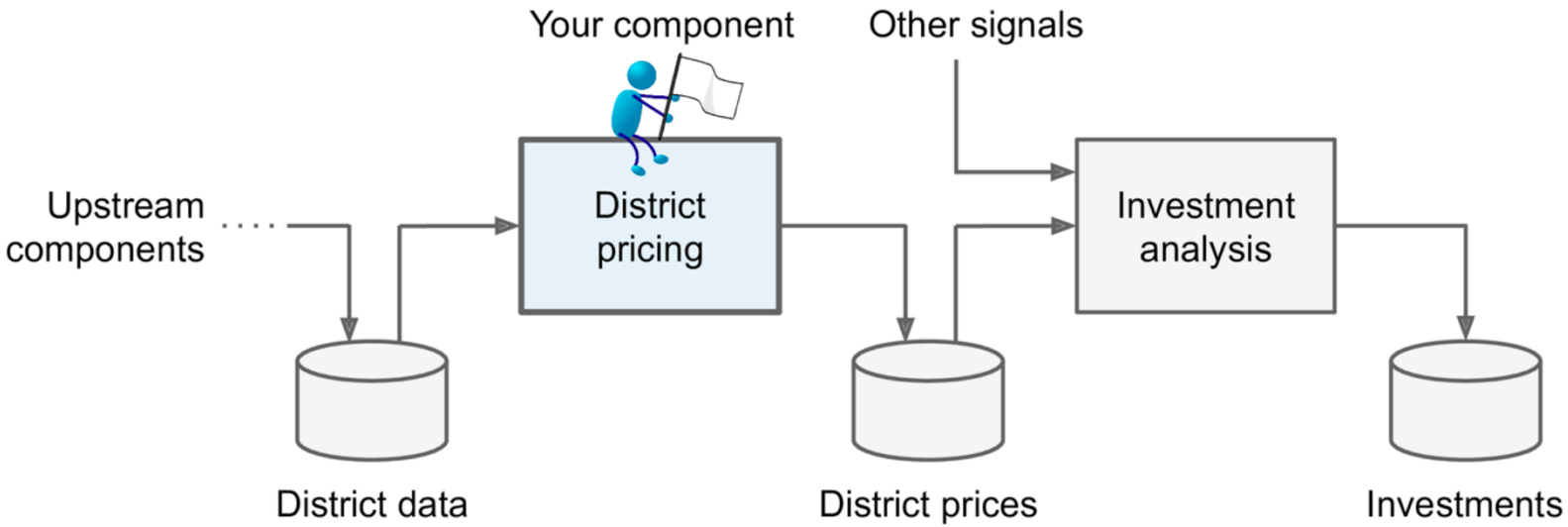


Figure 2-2. A Machine Learning pipeline for real estate investments

Real estate price prediction

- First, we need to frame the problem:
  - Is it supervised, unsupervised, or Reinforcement Learning?
  - Is it a classification task, a regression task, or something else?
  - Should we use batch learning or online learning techniques?

Performance measure

- Our next step is to select a performance measure.
- A typical performance measure for regression problems is the Root Mean Square Error (RMSE).
- It gives an idea of how much error the system typically makes in its predictions, with a higher weight for large errors.

$$\mathrm{RMSE}\left(\mathbf{X}, h\right) = \sqrt{\frac{1}{m}\sum_{i=1}^{m}\left(h\left(\mathbf{x}^{(i)}\right) - y^{(i)}\right)^2}$$

Performance measure

- m is the number of instances in the dataset you are measuring the RMSE on.
  - For example, if you are evaluating the RMSE on a validation set of 2,000 districts, then m = 2,000.

$$\mathrm{RMSE}\left(\mathbf{X}, h\right) = \sqrt{\frac{1}{m}\sum_{i=1}^{m}\left(h\left(\mathbf{x}^{(i)}\right) - y^{(i)}\right)^2}$$

Performance measure

- $\mathbf{x}^{(i)}$ is a vector of all the feature values (excluding the label) of the i-th instance in the dataset, and $y^{(i)}$ is its label (the desired output value for that instance).
  - For example, if the first district in the dataset is located at longitude –118.29° , latitude 33.91° , and it has 1,416 inhabitants with a median income of $38,372, and the median house value is $156,400 (ignoring the other features for now), then:

$$\mathbf{x}^{(1)} = \begin{pmatrix} -118.29 \\ 33.91 \\ 1{,}416 \\ 38{,}372 \end{pmatrix} \qquad y^{(1)} = 156{,}400$$

$$\mathrm{RMSE}\left(\mathbf{X}, h\right) = \sqrt{\frac{1}{m}\sum_{i=1}^{m}\left(h\left(\mathbf{x}^{(i)}\right) - y^{(i)}\right)^2}$$

Performance measure

- **X** is a matrix containing all the feature values (excluding labels) of all instances in the dataset.
  - There is one row per instance, and the i-th row is equal to the transpose of $\mathbf{x}^{(i)}$, noted $(\mathbf{x}^{(i)})^{\mathsf{T}}$.

$$\mathbf{X} = \begin{pmatrix} \left(\mathbf{x}^{(1)}\right)^{\mathsf{T}} \\ \left(\mathbf{x}^{(2)}\right)^{\mathsf{T}} \\ \vdots \\ \left(\mathbf{x}^{(1999)}\right)^{\mathsf{T}} \\ \left(\mathbf{x}^{(2000)}\right)^{\mathsf{T}} \end{pmatrix} = \begin{pmatrix} -118.29 & 33.91 & 1{,}416 & 38{,}372 \\ \vdots & \vdots & \vdots & \vdots \end{pmatrix}$$

$$\mathrm{RMSE}\left(\mathbf{X}, h\right) = \sqrt{\frac{1}{m} \sum_{i=1}^{m} \left(h\left(\mathbf{x}^{(i)}\right) - y^{(i)}\right)^2}$$

Performance measure

- h is our system's prediction function, also called a hypothesis.
  - When the system is given an instance's feature vector $\mathbf{x}^{(i)}$, it outputs a predicted value $\hat{y}^{(i)} = h(\mathbf{x}^{(i)})$.
  - For example, if the system predicts that the median housing price in the first district is \$158,400, then $\hat{y}^{(1)} = h(\mathbf{x}^{(1)}) = 158{,}400$. The prediction error for this district is $\hat{y}^{(1)} - y^{(1)} = 2{,}000$.

$$\mathrm{RMSE}(\mathbf{X}, h) = \sqrt{\frac{1}{m}\sum_{i=1}^{m}\left(h\left(\mathbf{x}^{(i)}\right) - y^{(i)}\right)^2}$$

Performance measure

- Convention:
  - We use lowercase italic font for scalar values (such as m or y) and function names (such as h), lowercase bold font for vectors (such as $\mathbf{x}^{(1)}$), and uppercase bold font for matrices (such as $\mathbf{X}$).

Performance measure

- Even though the RMSE is generally the preferred performance measure for regression tasks, in some contexts you may prefer to use another function.

Euclidean norm

$$\mathrm{RMSE}(\mathbf{X}, h) = \sqrt{\frac{1}{m}\sum_{i=1}^{m}\left(h\left(\mathbf{x}^{(i)}\right) - y^{(i)}\right)^2}$$

Manhattan norm

$$\mathrm{MAE}(\mathbf{X}, h) = \frac{1}{m}\sum_{i=1}^{m}\left|h\left(\mathbf{x}^{(i)}\right) - y^{(i)}\right|$$

More generally, the ℓ_k norm of a vector v containing n elements is defined as $\|v\|^k = (|v_0|^k + |v_1|^k + \cdots + |v_n|^k)$.

Performance measure

- Even though the RMSE is generally the preferred performance measure for regression tasks, in some contexts you may prefer to use another function.
- For example, suppose that there are many outlier districts. In that case, you may consider using the mean absolute error (MAE, also called the average absolute deviation)

Euclidean norm

$$\mathrm{RMSE}(\mathbf{X}, h) = \sqrt{\frac{1}{m}\sum_{i=1}^{m}\left(h\left(\mathbf{x}^{(i)}\right) - y^{(i)}\right)^2}$$

Manhattan norm

$$\mathrm{MAE}(\mathbf{X}, h) = \frac{1}{m}\sum_{i=1}^{m}\left|h\left(\mathbf{x}^{(i)}\right) - y^{(i)}\right|$$

More generally, the ℓ_k norm of a vector v containing n elements is defined as $\|v\|^k = (|v_0|^k + |v_1|^k + \cdots + |v_n|^k)$.

Performance measure

- The higher the norm index, the more it focuses on large values and neglects small ones. This is why the RMSE is more sensitive to outliers than the MAE.
- But when outliers are exponentially rare (like in a bell-shaped curve), the RMSE performs very well and is generally preferred.

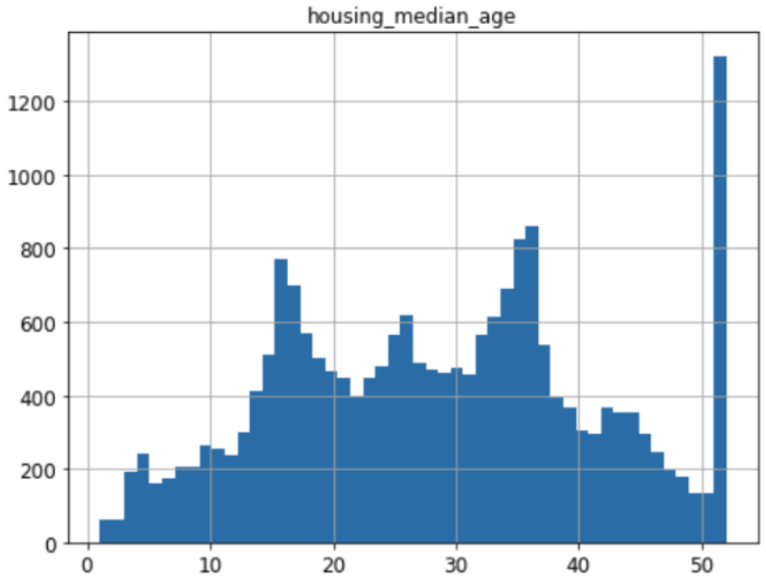
Euclidean norm

$$\mathrm{RMSE}(\mathbf{X}, h) = \sqrt{\frac{1}{m}\sum_{i=1}^{m}\left(h\left(\mathbf{x}^{(i)}\right) - y^{(i)}\right)^2}$$

Manhattan norm

$$\mathrm{MAE}(\mathbf{X}, h) = \frac{1}{m}\sum_{i=1}^{m}\left|h\left(\mathbf{x}^{(i)}\right) - y^{(i)}\right|$$

You



Describe this plot

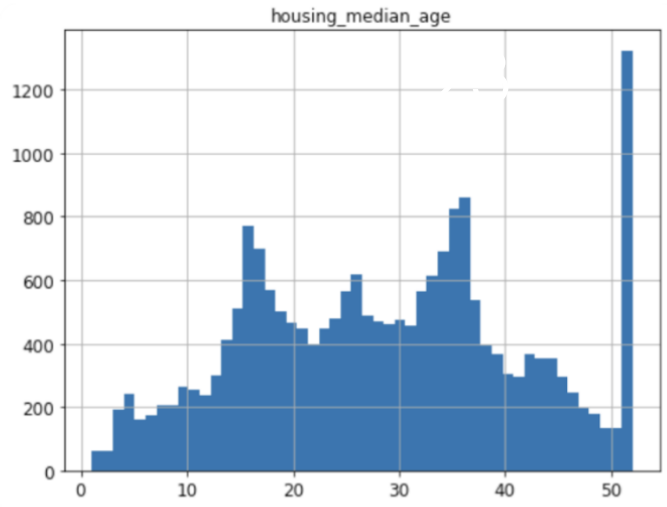
ChatGPT

The image you've uploaded is a histogram, which is a type of plot that shows the distribution of a dataset. This particular histogram appears to be displaying the variable 'housing_median_age'. The x-axis represents the median age of housing, segmented into bins, and the y-axis represents the frequency or count of occurrences within each bin.

The plot shows a distribution with several peaks, indicating that the median age of houses is not evenly distributed but rather clusters around certain values. There is a noticeable peak at the far right of the plot for the highest age bin, suggesting that there is a significant number of houses with a median age around 50 years. This could imply that there was a period of time when house building was very popular, resulting in a large number of houses that are now around the same median age.

The color used for the bars is blue, and there are no labels or legends that provide additional information about the data. To fully interpret this plot, one would need more context about the data source and the exact definitions of the age bins.

ChatGPT 5 Thinking



Describe this plot

Thought for 14s

It's a histogram of **housing_median_age** (in years).

- Ages range roughly **1–52**, with most tracts between **~15 and ~40** and several small bumps (mid-teens, late-20s, mid-30s).
- The standout feature is the **very tall bar at 52**, the maximum value—this indicates **censoring/clipping** in the data (many records are "52 or older" recorded as 52).
- Aside from that cap, the distribution is fairly spread out and somewhat multi-modal rather than a single bell curve.

Implications: consider treating 52 as "52+" (e.g., add an `is_capped` flag, or bucket ages) so that spike doesn't distort summaries or models.

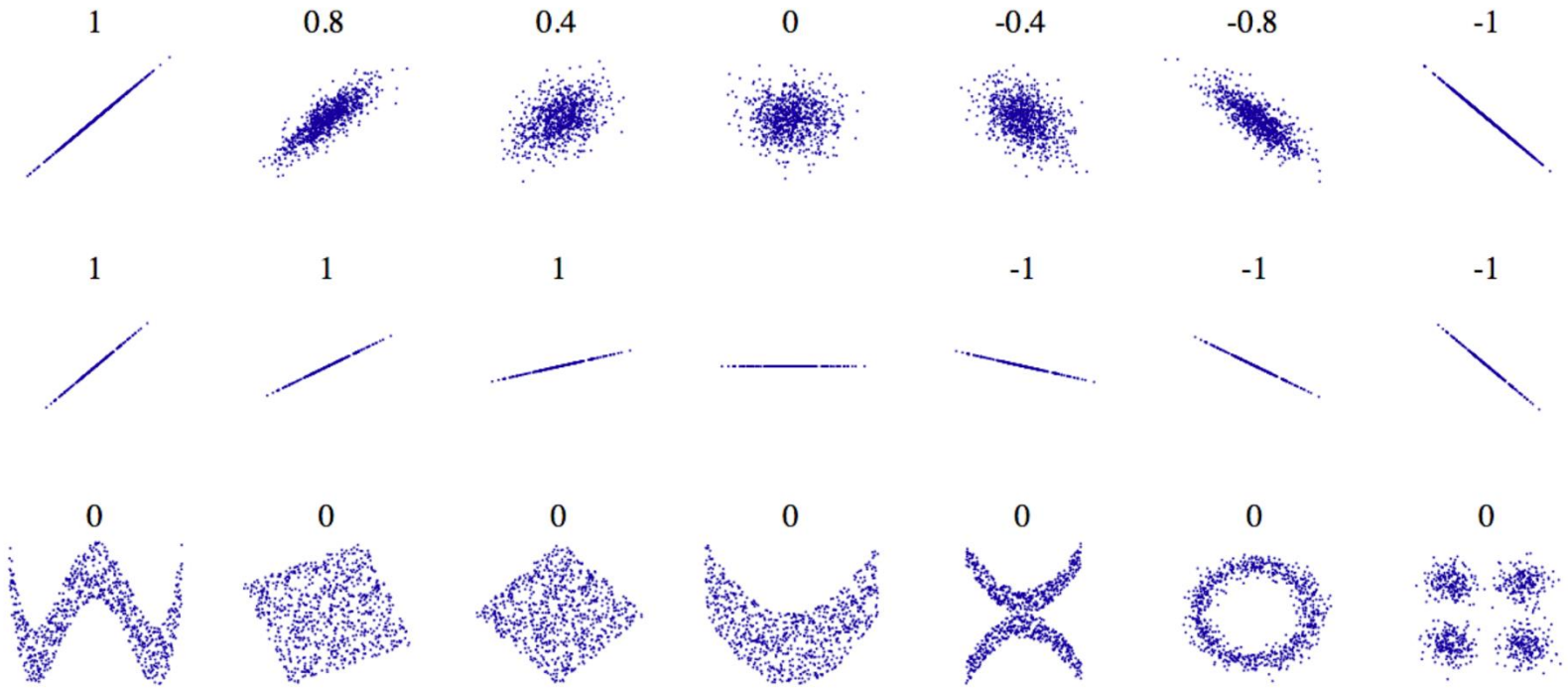


Figure 2-14. Standard correlation coefficient of various datasets (source: Wikipedia; public domain image)

WARNING

The correlation coefficient only measures linear correlations ("if x goes up, then y generally goes up/down"). It may completely miss out on nonlinear relationships (e.g., "if x is close to 0, then y generally goes up"). Note how all the plots of the bottom row have a correlation coefficient equal to 0, despite the fact that their axes are clearly not independent: these are examples of nonlinear relationships. Also, the second row shows examples where the correlation coefficient is equal to 1 or –1; notice that this has nothing to do with the slope. For example, your height in inches has a correlation coefficient of 1 with your height in feet or in nanometers.

Feature Scaling

- One of the most important transformations you need to apply to your data is feature scaling.
- There are two common ways to get all attributes to have the same scale: **minmax scaling** and **standardization**.
  - Min-max scaling (many people call this normalization) is the simplest: values are shifted and rescaled so that they end up ranging from 0 to 1. We do this by subtracting the min value and dividing by the max minus the min.

Min-max scaling (normalization) Example

$$x' = \frac{x - \min(x)}{\max(x) - \min(x)}$$

Example data: 10, 16, 22, 34, 40
min=10, max=40

Original x	$x - \min$	Scaled $x' = (x - 10)/30$
10	0	**0.00**
16	6	**0.20**
22	12	**0.40**
34	24	**0.80**
40	30	**1.00**

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  - Standardization is different: first it subtracts the mean value (so standardized values always have a zero mean), and then it divides by the standard deviation so that the resulting distribution has unit variance.

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Standardization (z score) Example

$$x' = \frac{x - \mu}{\sigma}$$ *(aka the z-score)*

Example data: 10, 16, 22, 34, 40

$$\mu = \frac{10 + 16 + 22 + 34 + 40}{5} = 24.4$$

Sample stdev: $\sigma = \sqrt{\frac{\sum(x-\mu)^2}{5-1}} = \sqrt{154.8} \approx 12.44$

Original x	$x - \mu$	Standardized $x' = (x - 24.4)/12.44$
10	-14.4	**-1.16**
16	-8.4	**-0.68**
22	-2.4	**-0.19**
34	9.6	**0.77**
40	15.6	**1.25**

Feature Scaling

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  - Standardization is different: first it subtracts the mean value (so standardized values always have a zero mean), and then it divides by the standard deviation so that the resulting distribution has unit variance.
  - Standardization is much less affected by outliers. For example, suppose a district had a median income equal to 100 (by mistake). Min-max scaling would then crush all the other values from 0-15 down to 0–0.15, whereas standardization would not be much affected.

Next time

- Intro to classification, based on Chapter 3 from *Hands on machine learning textbook*